EBU6304 Software Engineering Group Project

2025-26

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First report

1. Requirements

1.1 Fact-Finding Techniques

1.1.1 Background Reading

Background reading was one of the main fact-finding techniques used in the early stage of this project, because it helped us build an initial and high-level understanding of the system before carrying out more direct investigation.

(1) Understanding the project context

The project brief states that the current TA recruitment process at BUPT International School mainly relies on forms and Excel files, and that the new system should support functions such as applicant profile creation, CV upload, job search, job application, application status tracking, job posting by Module Organisers, applicant selection, and workload checking by administrators.

This showed that the project is not a general employment platform, but a recruitment system designed for a specific academic environment. It also made clear that the system would involve multiple user roles.

(2) Reviewing common functions in recruitment software

We also reviewed mainstream recruitment software to identify the core functions commonly found in applicant tracking systems.

Official materials from SAP, Workable, and Greenhouse show that such systems usually act as a central place for job postings and applicant information, and support activities such as posting positions, collecting applications, screening candidates, and managing candidate profiles.

Based on this review, we concluded that a lightweight recruitment system should normally include:

- publishing available positions;
- allowing applicants to view jobs and submit applications;
- storing applicant profiles and CVs;
- tracking application status;
- supporting staff-side review and selection.

(3) Understanding the real responsibilities of TAs in universities

Background reading was also used to understand the actual role of TAs in universities. Official university guidance from The University of Wisconsin–Milwaukee, Xi'an Jiaotong-Liverpool University, and The University of Texas at Austin shows that TA duties commonly include classroom support, leading discussion or lab sessions, holding office hours, grading assignments, giving feedback, proctoring exams, and assisting with course administration.

These examples suggest that different modules may require different numbers of TAs and different types of support depending on teaching format and workload. Therefore, the system should not only record whether a student applied for a role, but should also include information useful for assignment and workload coordination, such as role type, task requirements, skills, and availability.

1.1.2 Observation

Observation was another fact-finding technique used in this project. After building an initial understanding through background reading, we used observation to examine how TAs work in the school context, including how many TAs may be needed for one class and what tasks they support at different teaching stages.

Through observation, we found that TA work is not limited to a single activity. TAs may support classroom activities, practical or discussion sessions, student questions, marking, and examination-related tasks. We also observed that TA demand may vary depending on the module, class size, and teaching format.

This showed that TA recruitment is closely related to teaching needs and workload allocation, rather than simply filling a vacancy. Therefore, the system should not only support job application, but also provide clearer job descriptions, responsibility information, and better support for TA allocation and workload management.

1.1.3 Questionnaire

Questionnaires were used after background reading and observation to collect feedback from three stakeholder groups: TAs, MOs, and administrators. The aim was to identify key problems in the current recruitment process, understand which functions users valued most, and support later requirements refinement.

(1) TA perspective

For the TA questionnaire, most respondents had direct experience with the current process. Among 8 valid responses, 50% had applied for a TA position multiple times, 37.5% had applied once, and 12.5% had not yet applied but intended to do so in the future.

The main problems identified were the inability to track application status, difficulty uploading and managing CVs, and difficulty finding all available vacancies because job information was scattered. In the feature rating section, application status tracking, online CV upload, and centralised job search were rated as the most important functions. Respondents also wanted clearer job information, such as the MO, TA responsibilities, class size, and marking workload.

(2) MO perspective

The MO questionnaire reflected recruiter-side needs. The results suggested that the current process makes it harder to post jobs efficiently, review applicants consistently, and judge whether applicants match the required skills. It also showed that knowing whether students already hold other TA roles is useful for avoiding workload or timetable conflicts. In addition, there was a suggestion to support exporting the final TA list in CSV format.

(3) Administrator perspective

The administrator questionnaire focused on management-side concerns. The results showed that the most important needs were having a global overview of unfilled modules, tracking whether individual TAs were overloaded, and exporting final hiring results more easily. Respondents also rated workload transfer and balancing functions highly. This suggests that the system should support more centralised, transparent, and manageable TA allocation.

Overall, the questionnaire findings confirmed the importance of centralised job information, easier CV handling, application status tracking, clearer job descriptions, and better workload management.

1.2 Iteration plan

1.2.1 Iteration 1

In Iteration 1, the main work was to understand the problem, define the system, and build the initial working version.

We carried out fact-finding, identified the main user roles, analysed the current recruitment workflow, and converted the findings into user stories. These were then organised into a product backlog with priorities and estimates, and used to support the initial prototype design. Based on this, we selected a small set of core functions for implementation, including the basic interface, file-based data handling, and the main workflow for job browsing, application

submission, application status tracking, job posting, applicant selection, and checking TA overall workload.

1.2.2 Iteration 2

In Iteration 2, the focus was on extending the first version with additional support functions.

The team planned to add features such as viewing course and MO details, calculating working hours, a global vacancy dashboard, exporting the final hiring list, and selected AI-supported functions, including skill matching and missing-skill identification. The aim was to make the system more informative and useful for TA, MO, and admin users.

1.2.3 Iteration 3

In Iteration 3, the focus shifted to communication and management support functions.

The team planned to add online communication, a unified applicant list for MOs, and AI-supported workload balancing suggestions for administrators. These features were intended to improve coordination, applicant management, and allocation support.

1.2.4 Iteration 4

In Iteration 4, the focus was on completing the remaining lower-priority functions and polishing the system.

This included features such as leave requests and managing overloaded TAs, together with improvements to error handling, usability details, and overall system consistency. The overall priority remained delivering a stable system that satisfied the main customer needs.

1.3 Prioritisation methods

For prioritisation, our group used the MoSCoW method to organise the user stories in the product backlog. The product backlog stories are all prioritised according to business value, return on investment, and risk.

In practice, we reviewed the user stories one by one as a group and discussed three questions for each story: whether the function was necessary for the basic recruitment workflow, whether it was important to the main stakeholders, and whether it was realistic to implement within the project scope and technical constraints.

After this discussion, we classified the stories using **MoSCoW**:

- Stories were marked as **Must Have** if the system could not complete the core workflow without them. In our project, these mainly included creating and managing TA profiles, uploading and managing résumés, searching TA jobs centrally, applying for positions, tracking application status, posting job vacancies, selecting applicants, and checking the overall TA workload.

- Stories were marked as **Should Have** if they were important but not strictly required for the first working version. These mainly included viewing course and MO details, working-hour calculation, exporting the final hiring list, and some AI-supported functions such as skill matching and missing-skill identification.

- Stories were marked as **Could Have** if they mainly enhanced the system. These included functions such as online communication, viewing applicant lists, leave requests, and AI-supported workload balancing suggestions.

This follows the course explanation that Must have features are essential, Should have features are important but can be delayed, and Could have features are desirable enhancements.

1.4 Estimation methods

For estimation, our group used story points rather than exact working hours. This approach was more practical for our project because, in the early stages, it was difficult to predict precise development time for each user story.

In practice, we compared stories relative to one another. After the user stories had been prioritised, the team discussed each story in terms of several factors: the amount of interface work required, the complexity of the logic, the difficulty of file handling, the amount of testing needed, and the uncertainty of the requirements. This allowed us to judge whether one story was smaller, similar, or much larger than another.

To make the process more consistent, we used a simple reference-based approach:

- Small and simple stories, such as viewing course and MO details, exporting the final hiring list, or requesting leave, were given low points such as **2**.

- Stories with moderate interaction, such as managing TA profiles, uploading résumés, applying for jobs, posting vacancies, selecting applicants, or checking overall TA workload, were usually given **3 points**.

- Stories involving broader workflow or more complex logic, such as searching TA jobs, working-hour calculation, online communication, viewing applicant lists, or AI-supported functions like skill matching, missing-skill identification, and workload balancing suggestions, were given higher points such as **5**.

- More complex stories, such as tracking application status, were given **8 points** because they involved greater development effort and uncertainty.

The final point value for each story was decided through team discussion. After that, the story points were used together with the priority labels to support iteration planning, so that higher-priority and lower-effort stories could be placed earlier, while larger or less urgent stories were arranged later.

2 Supporting material

2.1 Background Research

2.1.1 Support material on common functions in recruitment software

(1) Source: SAP (https://www.sap.com/products/hcm/recruiting-software/what-is-an-applicant-tracking-system.html)

SAP

/ Human capital management / SmartRecruiters for SAP SuccessFactors

What is ATS?

What is an applicant tracking system (ATS)?

ATS is a software application for human resources (HR) that helps organizations manage recruitment and hiring tasks digitally.

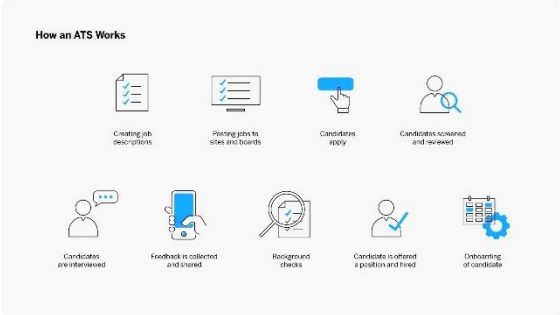
Explore ATS recruiting solutions

Applicant tracking system overview

An applicant tracking system (ATS) is a software application for human resources (HR) that helps organizations manage recruitment and hiring tasks digitally. It serves as the central database for job and applicant information where job requisitions, postings, and applications are managed and applicants are screened and selected.

An ATS supports the performance of common tasks such as creating job descriptions, posting jobs to career sites and job boards, résumé scanning, scheduling interviews, gathering feedback, performing background checks, and generating offer letters with electronic signatures.

The system also manages the experience for job applicants. This includes searching for jobs, applying online, and checking the status of an application, as well as generating notifications to applicants.



What are the capabilities of an end-to-end recruiting platform that includes ATS and CRM?

Here is an overview of the combined features and functions you can leverage with a comprehensive recruiting solution:

Posting a position

- Create job descriptions that emphasize preferred competencies.
- Identify gender bias and salary benchmarks using automated scans.
- Create an employer-branded website for job seekers.
- Build and manage postings on career websites and job boards.
- Use configurable applications per requisition (mobile-apply, quick-apply, multi-stage).

Reviewing applicants

- Search external and internal applications using a single database.
- Configure pre-application screening questions for each posting.
- Automate the storage and analysis of résumé data, including parsing.
- Receive automated notifications and alerts personalized for each role.
- Provide secure access for third-party agencies.
- Generate real-time reports and analytics across the entire recruitment process.

Interviewing candidates

- Schedule interviews and track progression.
- Gather feedback and automatically consolidate candidate ratings.
- Provides secure access to a candidate portal for current information and status.

Hiring and onboarding

- Access branded templates and forms with pre-filled data.
- Generate and present offer letters with electronic signature capabilities for faster turnaround.
- Automate the onboarding process.

(2) Source: workable (https://www.workable.com/features)

Applicant tracking system | ATS

CANDIDATE EVALUATION

Talent CRM database
Manage all applicants in one searchable database, with easy filtering and organization of candidate information.

Anonymized screening
Hide identifying candidate details during initial resume review to promote fair evaluation based on qualifications.

Standardized interview tools
Use pre-set and customizable interview kits and scorecards to ensure consistent candidate assessment across hiring teams.

AI-powered candidate screening
Get AI-generated summaries of candidate profiles, including match scores and requirement checklists.

One-way video interviews
Conduct one-way video interviews to efficiently screen candidates at scale, replacing traditional phone screenings.

Science-backed assessments
Administer science-backed cognitive and personality tests directly through the Workable platform.

REPORTING

Pre-built reports
Instant insights with ready-made reports for pipeline performance, time-to-hire, diversity, and source effectiveness - fully editable to match your workflow.

Custom report builder
Create reports from scratch using any field or metric across recruiting and HR. Combine tables, charts, and KPIs to track exactly what matters.

Access all data fields
Report on every available field - jobs, candidates, offers, interviews, referrals, and more - for complete visibility across your data.

Filter by any metric or dimension
Slice results by team, department, location, job type, or source to uncover trends and spot bottlenecks instantly.

Customizable dashboards
Build dashboards with your saved reports, rearrange charts and tables, and give every stakeholder a view tailored to their goals.

Create and save your own reports
Save custom configurations as reusable templates and quickly reload them to compare time periods, regions, or business units.

Export your reports
Export reports to Excel, CSV, or PDF, or share them directly inside Workable with role-based access and scheduled email delivery.

Automated report emails
Send dashboards or reports automatically on your chosen cadence - daily, weekly, or monthly - to keep stakeholders informed without manual effort.

BI-data link
Connect Workable data to your organization's BI tools for deeper analytics and cross-system reporting.

AUTOMATION AND HIRING

Automated recruitment tasks
Create custom email templates and workflows to automate repetitive tasks based on job, department, or hiring stage.

Candidate text messaging
Communicate with candidates via SMS messages for quick updates and a modern candidate experience.

Advanced reporting and analytics
Generate automated reports from a vast range of customizable, pre-built reports, and integrate with BI tools for in-depth analysis.

Offer letters and approval workflows
Speed up sign-off and get signatures faster with offer letter templates, approval workflows, e-signatures and more.

Custom hiring pipelines
Create customized hiring workflows to match your specific recruitment needs and processes for different roles.

Department hierarchy
Organize jobs and tools by department to match company structure and ensure the right teams follow the right process.

GDPR, CCPA and EEOC/OFCCP compliance
Use automated tools and reports to maintain compliance with GDPR, CCPA, EEOC/OFCCP, and other regulations across jurisdictions.

Candidate surveys
Gather insights on candidate experience using customizable survey forms sent to applicants at various stages.

Developer tool access
Utilize the Developer API and connect Workable with your tools for custom integrations and analysis. Or have our Professional Services team do it for you.

Requisition and budgeting workflows
Keep stakeholders aligned with a centralized workspace for budgeting, hiring and requisition approval workflow management.

Salary estimator
Access accurate market salary data to inform your hiring and budgeting decisions effortlessly (additional cost applies).

COLLABORATION TOOLS

2-way email and calendar integration
Sync with email and calendar apps such as Google and Microsoft Outlook to manage candidate communications and scheduling.

Video conferencing integrations
Set up virtual interviews easily with Workable's built-in Zoom, Google Meet, and Microsoft Teams integrations.

Third-party tool integrations
We integrate with 70+ recruiting and HR tools, so you can assess, background check and onboard your next great employee, all in Workable.

Candidate profile sharing
Share candidate profiles with team members to gather feedback, facilitate discussions, and prepare for interviews collaboratively.

Customizable user permissions
Set specific access rights and roles for hiring team members to manage confidentiality and task allocation effectively.

Customizable candidate profiles
Add custom fields to candidate profiles to track relevant information while protecting sensitive data like salary details.

Mobile app for hiring (and HR) teams
Available for iOS and Android, Workable's mobile app helps teams keep hiring moving, even when they're not behind a desk.

Candidate self-scheduling
Empower candidates to book interview slots directly. Reduce back-and-forth emails with automatic calendar syncing and time zone detection.

Comprehensive interview coordination
Set up multi-part interviews, add custom event details, and provide location maps. Sync with team calendars as events are booked and changed.

ACCESS AND PERMISSIONS

Preset roles for Recruiting and HR
Workable comes with predefined roles for Recruiting and HR teams - like Recruiter, Hiring Manager, Contributor, and Reviewer - each with built-in permissions that fit common hiring workflows.

Customizable roles and permissions
Adapt default roles to your process by applying additional permission sets. Choose exactly what people can see and do, from editing jobs to viewing reports, for precise control without complexity.

Advanced permission sets
Layer additional access controls on top of roles to support specialized needs - such as Finance, IT, or regional leads - defining who can view, edit, comment, or approve.

Governed data sharing
On the Enterprise plan, when reports or dashboards are shared, Workable automatically applies role-based visibility so stakeholders only see permitted information.

Smart user access and invites
Invite new users in seconds or let colleagues join automatically using your company email domain. When added to a hiring team, accounts are created and given the right role for that job.

Single sign-on and provisioning
Secure your account with SSO, and for Enterprise customers, enable SCIM provisioning to sync users and deactivate accounts automatically.

Audit logging
Track every change to roles, reports, and permissions for full visibility and compliance peace of mind.

HR information system | HRIS

ONBOARDING

Centralized onboarding portal
Provide new hires with a single "home base" for welcome messages, company videos, e-signature documents, and personal profile completion.

Role-specific onboarding templates
Create and apply customized onboarding workflows for different departments, locations, or roles to ensure relevant experiences for each new hire.

Simplified document management
Simplify the collection and organization of new hire paperwork with e-signature capabilities and secure document storage.

Custom onboarding tasks
Create custom tasks for new hires, line managers, and admins to complete during an employees onboarding.

Onboarding progress dashboard
Track the status of each new hire's onboarding journey, including start and end dates, completed tasks, and outstanding items.

Team collaboration tools
Keep your HR team and line managers informed with shared access to onboarding statuses, notifications, timelines, and task completion for each new

EMPLOYEE MANAGEMENT

Flexible employee profiles
Use customizable profile templates to collect employee information, using conditional field logic to show, hide, or populate fields dynamically.

Self-service employee records
Allow employees to update their own personal details with approval flows and change history tracking, reducing manual data change requests.

Organized, searchable company file storage
Store and manage company-wide documents and e-signed files securely with controlled access.

Fillable document templates with e-signatures
Create branded, fillable document templates for easy employee review and digital signing.

Automated org chart
Maintain an up-to-date visual representation of your company structure, automatically updated with personnel changes.

Searchable employee directory
Find and connect with any employee using search and filter options within the organization.

Employee lifecycle management
Manage the entire employee journey. Features include hiring, onboarding, data management, time off, performance reviews, and offboarding.

Multi-currency support
Handle multiple currencies for employee and candidate profile fields. Set up salary and compensation details in the appropriate currency for each location.

Comprehensive interview coordination
Set up multi-part interviews, add custom event details, and provide location maps. Sync with team calendars as events are booked and changed.

AI-powered HR tools

CREATING CONTENT

AI job description generator
Create role-specific job descriptions tailored to your company in seconds, resulting in higher application rates.

AI interview question generator
Generate industry and role-specific interview questions to ensure consistent and thorough candidate evaluations.

Personalized email creator
Craft emails that reference both candidate qualifications and job requirements, improving response rates.

SOURCING & MATCHING CANDIDATES

AI candidate sourcing
Use AI to scan our database of over 400 million passive candidate profiles to proactively find matches for your jobs before you even start searching.

Resurface candidates
AI automatically identifies and resurfaces qualified past candidates from your existing talent pool whenever you post new job openings.

Intelligent social targeting
With the click of a button, Workable's AI identifies 1000+ of your top passive candidates, then displays your job ad in their social media feeds.

AI-assisted applicant screening
Match candidates to job requirements using semantic analysis to see skill/experience matches easily, enabling faster and more accurate profile screening.

AUTOMATING TASKS

Intelligent resume parsing
Automatically extract and enhance key details from resumes, populating data fields accurately for unbiased screening.

Automated candidate outreach
Generate and send personalized outreach emails to potential candidates based on their qualifications and job requirements.

Real-time salary estimator
Access up-to-date salary information for any role to inform hiring plans and salary band decisions.

Workable Assistant
Get expert HR insights and guidance to help your team work smarter, every step of the way.

(3) Source: greenhouse (https://support.greenhouse.io/hc/en-us/articles/11957068130971-Using-the-new-candidate-profile)

Action	Steps
Find a candidate's resume	Click **Resume** on the right side of the header. You can also find the candidate's resume in the panel on the right side of the page under **Candidate** or **Application** > **Documents**.
Find candidate notes	Click the **Notes** icon on the right side of the page.
See the full list of jobs a candidate is on	Click the **Candidate** icon on the right side of the page, then scroll down to **All Jobs**.
Determine if the candidate is a duplicate	Click the **Alerts** icon. If the candidate is a potential duplicate, an error message will display.
Navigate between candidates on the same job	Use the arrows in the top right corner of the candidate profile.
Go to a job's dashboard	From the navigational links in the top left corner, click the job name. If you've recently viewed the job, you can also click on the job name in the **Recently Viewed** menu in the bottom right corner of any page.
View all jobs for a candidate	Open the tab on the right panel

2.1.2 Support material on the real responsibilities of TAs in universities

(1) **Source: The University of Wisconsin–Milwaukee (https://uwm.edu/graduate-assistants/handbook/teaching-assistants/roles-and-responsibilities-of-teaching-assistants/)**

Roles and Responsibilities of Teaching Assistants

The main role of a teaching assistant is to provide support to the course instructor to ensure the effective delivery of the required materials and to foster a positive learning environment for students. Given the vast array of possible courses requiring Teaching Assistants at UWM, their individual roles will vary considerably. Each assistantship is tailored to the needs of each specific course and determined by the instructor.

The following is a list of some typical responsibilities that teaching assistants may be expected to carry out. It is designed to help instructors devise their own guidelines to suit the needs of their specific course. Regular communication and feedback between instructors and Teaching Assistants is of paramount importance to ensure expectations are met both by the instructor and the TA. The Teaching Assistant should meet with the instructor frequently throughout the semester to discuss class materials and duties.

The responsibilities of Teaching Assistants are highly variable, but can include:

- Assisting faculty with classroom instruction, records, and assignments
- Leading discussion sections
- Meeting with students during office hours
- Conferencing with students individually or in small groups
- Delivering lectures or guest lectures
- Leading group projects or discussions
- Grading assignments or papers
- Managing course communications in Canvas
- Preparing laboratory materials
- Recording and calculating grades
- Providing feedback on assignments
- Enforcing laboratory rules and procedures
- Proctoring examinations
- Taking attendance or monitoring participation
- Obtaining and distributing course materials
- Ordering course textbooks and monitoring supply

(2) Source: Xi'an Jiaotong-Liverpool University (https://www.xjtlu.edu.cn/en/teaching-assistants)

To meet the teaching demands for the upcoming semester, Xi'an Jiaotong-Liverpool University (XJTLU) is now recruiting Teaching Assistants (TAs) across various Schools/Academies. Detailed job postings and application instructions are available on the recruitment page of each School/Academy, listed as follows.

TA duties include:

- Laboratory demonstration and support for practical work in the classroom;
- Attendance on and support with field courses;
- Group tutoring;
- Scheduled office hours for one-to-one tutoring;
- Invigilation of formal examinations and/or class tests;
- Marking of formative assignments with appropriate training; and
- Other appropriate activities as determined by academic units.

Specific responsibilities are detailed in the TA advertisements posted on each School/Academy page as follows.

Skills and knowledge required:

- Proficient spoken and written English;
- Excellent communication and interpersonal skills;
- Excellent presentation skills, and skillful in Microsoft Office Software (Word, Excel, PowerPoint, etc.);
- Excellent organizational skills and attention to detail;
- Be adaptable and open to change; and
- Previous TA working experience preferred.

Please refer to each School/Academy's TA advertisements for detailed requirements.

TA training:

The XJTLU Graduate School will coordinate with the Educational Development Unit (EDU) of the Academy of Future Education to offer a series of TA training sessions to all incoming TAs. These sessions aim to equip students with essential teaching theories, methods, and practical skills. Upon completion of all training sessions, participants will receive a TA Training Certificate, enabling them to officially take on TA roles. This certificate can also enhance students' future career development.

(3) Source: The University of Texas at Austin (https://gradschool.utexas.edu/funding/student-employment/academic/work-agreement/TA)

Job description

HOP General Job Description

The Handbook of Operating Procedures (HOP) provides the general job description for teaching assistants (HOP 9-2020).

TAs are defined as "graduate students who help faculty with the conduct and delivery of courses. Services provided by TAs include, but are not limited to, grading, monitoring, leading lab and/or discussion sessions, offering office hour assistance to students, and performing clerical tasks associated with course instruction. TAs are supervised by the course instructor of record and are subject to established departmental policies on student academic employment."

TA-instructor agreements should clearly specify the expected duties of the TA for the assigned course, and these duties must be acknowledged and agreed to by both the instructor and the student. Checklists can be useful for this purpose, where the instructor selects the duties from a list of common duties. Examples of checklist duties include:

- Grade or otherwise evaluate student work
- Monitor and lead lab sections, student discussion sessions, recitation sessions, and similar activities
- Hold office hours for student assistance
- Perform clerical tasks associated with course instruction
- Proctor exams
- Provide feedback to the instructor about class performance

TAs are prohibited from regular classroom instruction and may not serve as the instructor of record. The TA-instructor agreement should be explicit about expectations for TA classroom instruction, including specifying circumstances when lecturing to the class is allowed (such as under the direct supervision of the instructor for mentoring/training purposes) and the frequency of such activities.

It is also good practice to specify in the agreements activities that are prohibited, such as performing personal tasks for the instructor. TAs and instructors should also agree upon methods for communicating during the semester and frequency of meetings, guidance and feedback. International students may have communication, guidance and feedback needs different from domestic students, so it is important that these expectations are discussed and agreed upon by the TA and instructor.

Center for Teaching and Learning

Students serving as TAs are encouraged to use the Center for Teaching and Learning's resources for training and support.

2.2 Questionnaires and results

2.2.1 TA Questionnaires and results

(1) Questionnaires

BUPT International School Teaching Assistant (TA) Recruitment System Requirements Survey

Hello! We are a software engineering development team. To provide you with a more convenient TA application experience, we invite you to participate in this anonymous requirements survey.

*1. Have you ever applied for a Teaching Assistant (TA) position at BUPT International School in the past

- ○ Yes, multiple times
- ○ Yes, once
- ○ No, but I intend to in the future

*2. In the traditional TA application process, what do you consider to be the most cumbersome aspects or pain points[Multiple]

- ☐ Unable to find all vacant positions (job information is scattered)
- ☐ Having to repeatedly fill out personal information for different modules
- ☐ Inability to conveniently upload and manage my CV/resume
- ☐ Unable to track my application status after applying (e.g., Passed/Rejected/Pending)
- ☐ Other (Please specify)

*3. If a brand-new TA recruitment system were to be implemented, please rate the importance of the following features to you (1 = Not needed at all, 5 = Extremely important)

	1 – Not needed at all	2	3	4	5 – Extremely important
Create an exclusive applicant profile	○	○	○	○	○
Upload CV/resume online	○	○	○	○	○
Centrally search for all currently available TA jobs	○	○	○	○	○
One-click job application	○	○	○	○	○
Check real-time application progress/status for each job	○	○	○	○	○

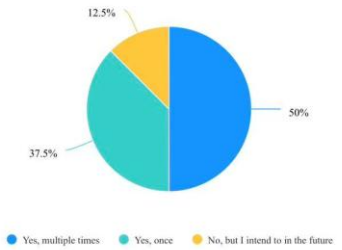
*4. Besides the basic features mentioned above, what other features or assistance do you hope the system will provide for you as a TA

(2) Results

1.Have you ever applied for a Teaching Assistant (TA) position at BUPT International School in the past [Single Choice]

Option	Subtotal	Percentage
Yes, multiple times	4	50%
Yes, once	3	37.5%
No, but I intend to in the future	1	12.5%
Valid Count Per Participant	**8**	

Pie Chart | Donut Chart | Bar Chart | Bar | Line Chart



Numeric | Sorting

2.In the traditional TA application process, what do you consider to be the most cumbersome aspects or pain points [Multiple Choice]

Option	Subtotal	Percentage
Unable to find all vacant positions (job information is scattered)	2	25%
Having to repeatedly fill out personal information for different modules	0	0%
Inability to conveniently upload and manage my CV/resume	3	37.5%
Unable to track my application status after applying (e.g., Passed/Rejected/Pending)	7	87.5%
Other (Please specify) [Details]	0	0%
Valid Count Per Participant	**8**	

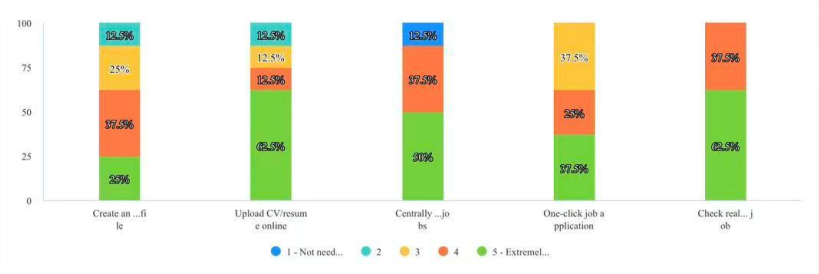
Pie Chart | Donut Chart | Bar Chart | Bar | Line Chart

4.If a brand-new TA recruitment system were to be implemented, please rate the importance of the following features to you (1 = Not needed at all, 5 = Extremely important) [Matrix Rating]

Average score of the matrix question: 4.15 View Details

	1 – Not needed at all	2	3	4	5 – Extremely important	Average Score
Create an exclusive applicant profile						
	0(0%)	1(12.5%)	2(25%)	3(37.5%)	2(25%)	3.75
Upload CV/resume online						
	0(0%)	1(12.5%)	1(12.5%)	1(12.5%)	5(62.5%)	4.25
Centrally search for all currently available TA jobs						
	1(12.5%)	0(0%)	0(0%)	3(37.5%)	4(50%)	4.13
One-click job application						
	0(0%)	0(0%)	3(37.5%)	2(25%)	3(37.5%)	4
Check real-time application progress/status for each job						
	0(0%)	0(0%)	0(0%)	3(37.5%)	5(62.5%)	4.63
Subtotal	1(2.5%)	2(5%)	6(15%)	12(30%)	19(47.5%)	4.15

Bar Chart | Bar | Line Chart | Parallel Coordinates | Radar Chart



5.Besides the basic features mentioned above, what other features or assistance do you hope the system will provide for you as a TA [Textbox]

Word Frequency Analysis | Opinion Analysis | AI Opinion Analysis | Hide Word Cloud | Details

2.2.2 MO Questionnaires and results

(1) Questionnaires

BUPT International School TA Recruitment System MO Questionnaire

Dear School Module Organizers, thank you for participating in this survey. Your feedback is crucial forimproving the TA recruitment coordination system. Please answer the following questions based on youractual experience and needs.

*1. On average, how many teaching assistants do you need to recruit per semester to support your module?

- ○ 1–2 TAs
- ○ 3–5 TAs
- ○ More than 5 TAs

*2. What are the main difficulties you encounter during the current TA recruitment and screening process? (Select all that apply)[Multiple]

- ☐ Posting job information is cumbersome and lacks a unified channel
- ☐ Received student CVs are in various formats, making quick screening difficult
- ☐ Hard to judge whether a student possesses the specific skills required for the module
- ☐ Difficult to track which students have already been hired by other modules (leading to scheduling/workload conflicts)
- ☐ Other (Please specify):

*3. For the new recruitment system, please rate the importance of the following features to you

	1–Not needed at all	2	3	4	5–Extremely important
One–click posting of job vacancies and skill requirements:	○	○	○	○	○
View all applicants in a unified list and select to accept/reject them:	○	○	○	○	○
[AI Feature] System automatically compares applicant skills with job requirements and provides a matching score:	○	○	○	○	○
[AI Feature] System automatically identifies and highlights missing key skills for applicants:	○	○	○	○	○

*4. [Open–ended] Regarding the features above or your daily TA recruitment tasks, are there any other features you expect the system to implement?

(2) Results

*1. On average, how many teaching assistants do you need to recruit per semester to support your module?

More than 5 TAs

*2. What are the main difficulties you encounter during the current TA recruitment and screenin g process? (Select all that apply) [多选题]

Posting job information is cumbersome and lacks a u nified channel ¦ Received student CVs are in various formats, making quick screening difficult ¦ Hard to jud ge whether a student possesses the specific skills re quired for the module ¦ Difficult to track which studen ts have already been hired by other modules (leading to scheduling/workload conflicts) ¦ Other (Please spe cify): 〖TA information is in a large file, I need to filter who applied my module. Their CV are in a separate f older and I have to find them manually.〗

*3. For the new recruitment system, please rate the importance of the following features to you [矩阵量表题]

One-click posting of job vacancies and skill requi rements:
5-Extremely important
5 Points

View all applicants in a unified list and select to a ccept/reject them:
5-Extremely important
5 Points

[AI Feature] System automatically compares appl icant skills with job requirements and provides a matching score:
3
3 Points

[AI Feature] System automatically identifies and highlights missing key skills for applicants:
2
2 Points

平均分：3.75

*4. [Open-ended] Regarding the features above or your daily TA recruitment tasks, are there an y other features you expect the system to impl ement?

As a MO, I want to have a final list of confirmed TA in my module, as a simple CSV file, so I can easily cop y the list to other places.

2.2.3 Admin Questionnaires and results

(1) Questionnaires

TA Recruitment Coordination System Requirements Survey (Admin)

Dear School Administrator, thank you for participating in this survey. Your feedback is crucial for improving the TA recruitment coordination system. Please answer the following questions based on your actual experience and needs.

*1. As a school administrator coordinating university-wide TA recruitment, what are your core demands or pain points[Multiple]

- [ ] Unable to have a global overview of which modules are still unfilled
- [ ] Difficult to track whether a single TA has taken on an excessive workload (e.g., holding positions in 3 modules simultaneously)
- [ ] Lack of data reports, unable to quickly export the final hiring list
- [ ] Other (Please specify)

*2. Please rate the importance of the following management features to you

	1 – Not needed at all	2	3	4	5 – Extremely important
View a global dashboard to monitor the overall workload of all TAs	○	○	○	○	○
Force dismiss or transfer positions for overloaded TAs	○	○	○	○	○
[AI Feature] System intelligently warns and provides workload balancing suggestions	○	○	○	○	○

*3. In terms of coordinating TA resources and workload, do you need any other macro-management features from the system

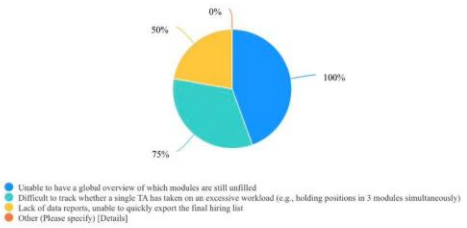
(2) Results

TA Recruitment Coordination System Requirements Survey（Admin）

1.As a school administrator coordinating university-wide TA recruitment, what are your core demands or pain points [Multiple Choice]

Option	Subtotal	Percentage
Unable to have a global overview of which modules are still unfilled	4	100%
Difficult to track whether a single TA has taken on an excessive workload (e.g., holding positions in 3 modules simultaneously)	3	75%
Lack of data reports, unable to quickly export the final hiring list	2	50%
Other (Please specify) [Details]	0	0%
Valid Count Per Participant	**4**	

Pie Chart | Donut Chart | Bar Chart | Bar | Line Chart



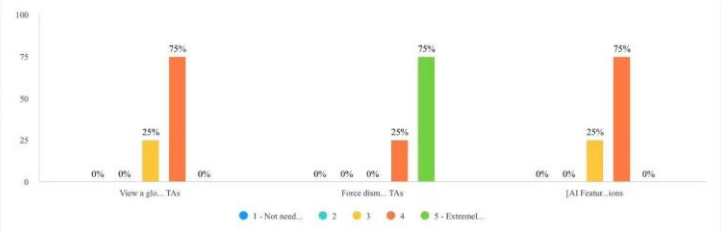
Numeric | Sorting

3.Please rate the importance of the following management features to you [Matrix Rating]

Average score of the matrix question: 4.08 View Details

	1 – Not needed at all	2	3	4	5 – Extremely important	Average Score
View a global dashboard to monitor the overall workload of all TAs						
	0(0%)	0(0%)	1(25%)	3(75%)	0(0%)	3.75
Force dismiss or transfer positions for overloaded TAs						
	0(0%)	0(0%)	0(0%)	1(25%)	3(75%)	4.75
[AI Feature] System intelligently warns and provides workload balancing suggestions						
	0(0%)	0(0%)	1(25%)	3(75%)	0(0%)	3.75
Subtotal	0(0%)	0(0%)	2(16.67%)	7(58.33%)	3(25%)	4.08

Bar Chart | Bar | Line Chart | Parallel Coordinates | Radar Chart



Numeric

序号	答案文本
1	I hope the system can clearly show who the MO is for each course, what the TA is expected to do, the class size, and whether the marking workload is heavy. It would also be helpful if the system could support communication with teachers or recruitment staff and provide a simple way to calculate working hours.
2	Historical allocation record queries，making it convenient to see which modules and positions a TA has undertaken in the past.
3	early warning of vacant positions to remind when TA is not recruited for a long time in some modules.
4	一键导出分配结果